Sample Project Documentation

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

Project Title

Personalized Travel Itinerary Planner

1. Project Overview

The Personalized Travel Itinerary Planner is a full-stack web application that leverages Generative AI (Google Gemini / Groq LLMs) to create customized, budget-aware, day-by-day travel itineraries. Users provide trip parameters such as destination, dates, budget, group size, travel style, and interests, and the system generates a complete travel plan with activities, timings, locations, and cost estimates.

2. Problem Statement

Travelers face significant challenges when planning personalized itineraries: information overload from multiple sources, difficulty managing budgets across activities, generic recommendations that don't match individual preferences, and rigid itineraries that can't be easily customized. This project solves these problems by providing AI-powered, personalized itinerary generation with full customization capabilities.

3. Technology Stack

- **Frontend**: React 19, TypeScript, Vite, TailwindCSS v4, shadcn/ui, React Query, React Hook Form, Zod, dnd-kit
- **Backend**: Python, FastAPI, Pydantic, PyJWT, HTTPX
- **Database**: Supabase (PostgreSQL with Row-Level Security)
- **AI**: Google Gemini API (primary), Groq API (fallback)
- **Deployment**: Vercel (frontend), Render (backend)

4. Architecture

The application follows a three-tier architecture:

- **Presentation Layer**: React SPA with client-side routing, form validation, and state management
- **Business Logic Layer**: FastAPI backend with service layer pattern (routers → services → repositories)
- **Data Layer**: Supabase PostgreSQL with RLS policies and 6 tables (profiles, trips, itinerary_days, activities, trip_members, generation_logs)

5. Key Features

1. AI-powered multi-day itinerary generation using Gemini/Groq LLMs
2. Secure authentication with Supabase Auth and JWT/JWKS verification

3. Full trip CRUD with budget tracking and cost estimation
4. Day-by-day itinerary view with activity details
5. Single-day regeneration without affecting other days
6. Drag-and-drop activity reordering with time slot swapping
7. Activity editing and deletion
8. Row-Level Security for complete data isolation
9. LLM fallback mechanism (Gemini → Groq)
10. Responsive UI with loading states and error handling

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6. API Documentation

The API is fully documented with interactive Swagger UI at:

https://personalized-travel-itenerary-planner.onrender.com/docs

Key Endpoints:

- POST /api/trips/generate — Generate AI itinerary
- GET /api/trips — List trips
- GET /api/trips/{id} — Get trip detail
- PATCH /api/trips/{id} — Update trip
- DELETE /api/trips/{id} — Delete trip
- POST /api/trips/{id}/days/{n}/regenerate — Regenerate day
- POST /api/trips/{id}/days/{n}/reorder — Reorder activities
- PATCH /api/activities/{id} — Update activity
- DELETE /api/activities/{id} — Delete activity

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7. Database Schema

- **profiles**: User profiles with display name and preferences
- **trips**: Trip metadata (destination, dates, budget, style, interests, status)
- **itinerary_days**: Days within a trip (day number, date, summary, cost)
- **activities**: Individual activities (type, title, description, location, times, cost)
- **trip_members**: Invited collaborators with roles (editor/viewer)
- **generation_logs**: AI generation audit trail

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8. Security Implementation

- Supabase Auth for email/password authentication
- JWT verification using JWKS with key caching (600s lifespan)
- 14 Row-Level Security policies across 6 tables
- SECURITY DEFINER helper functions for safe policy evaluation
- CORS restriction to frontend origin only
- Input validation on both frontend (Zod) and backend (Pydantic)

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9. Deployment

Component	Platform	URL
Frontend	Vercel	https://personalized-travel-itenerary-planner-ob42m5s4v-tanay-e4be.vercel.app/
Backend	Render	https://personalized-travel-itenerary-planner.onrender.com
API Docs	Swagger UI	https://personalized-travel-itenerary-planner.onrender.com/docs
Database	Supabase	Project ID: vskaghlmvjqambnrshhy
Source Code	GitHub	https://github.com/Tanay-Ds/Personalized-Travel-Itenerary-Planner